

STRUCTURAL AND THERMOTROPIC EFFECTS OF LIPID HYDROPEROXIDATION IN LIPID BILAYERS

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ABSTRACT

Lipid peroxidation is a defining feature of oxidative stress and a major factor in membrane dysfunction during aging, inflammation, and disease. While oxidation-induced changes in bilayer structure are well recognized (1), their impact on the main phase transition and membrane organization of phospholipids remains poorly understood (2). Here, we investigate how hydroperoxidation affects the phase behavior and structural properties of two phosphatidylcholine lipids, 1-stearoyl-2-oleoyl-sn-glycero-3-phosphocholine (SOPC) and 1-stearoyl-2-petroselinoyl-sn-glycero-3-phosphocholine (SPPC), together with their hydroperoxidized forms.

Mixtures of oxidized and non-oxidized lipids were characterized using complementary techniques. Differential scanning calorimetry (DSC) and steady-state fluorescence spectroscopy with the environment-sensitive probes Laurdan (3) and A10 (4) were used to probe phase behavior. Membrane structure and organization were further investigated using neutron scattering. Hydroperoxidation strongly alters membrane thermodynamics and nanoscale structure, revealing a complex phase-transition landscape driven by acyl-chain ordering and hydroperoxide distribution (5). Together, these measurements provide a multiscale view of how lipid peroxidation modulates membrane organization, offering insight into the molecular consequences of oxidative stress in biomimetic membranes.

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